

Welcome to MECH 191

Course Description

A comprehensive introduction to the principles of metallurgy and material science, focusing on the structure, properties, and performance of metals and alloys. Topics include mechanical testing, metallography, phase diagrams, heat treatment, mechanical properties, corrosion, and failure analysis.

You'll gain hands-on experience in material testing, alloy composition, and microstructural analysis, and learn how to select and apply materials for real industrial purposes.

Learning Objectives

- Explain the relationship between structure, properties, and performance of metals and alloys
- Describe the phases in a metal alloy system using a phase diagram
- Apply heat treatment processes to modify metal properties for engineering applications
- Conduct lab tests to measure hardness, tensile strength, and ductility
- Interpret failure mechanisms — corrosion and fatigue — from properties and usage conditions
- Assess metal/ally suitability for engineering scenarios: performance, durability, cost

SLIDE 2 *Welcome to MECH 191*

NOTES

Assignment	Count	Value Each	Total Points
Attendance	15	1	15
Metallography Portfolio	1	50	50
Unit Quizzes	5	5	25
Final Exam	1	10	10
Total			100

Must be returned by 7:00 PM or the score is automatically zero.

The Metallography Portfolio (50 pts) is graded internally across all 15 weekly modules plus a holistic end-of-semester review — see the Metallography Lab slide for details.

NOTES

Course Timeline

Wk	Date	Topic
1	Aug 24	Syllabus, PPE / Unit 1 begins
2	Aug 31	Unit 1 (cont.) — Unit 1 Quiz
3	Sep 7	NO CLASS
4	Sep 14	Unit 2 begins
5	Sep 21	Unit 2 (cont.)
6	Sep 28	Unit 2 (cont.) — Unit 2 Quiz
7	Oct 5	Unit 3 begins
8	Oct 12	Unit 3 (cont.)
9	Oct 19	Unit 3 (cont.) — Unit 3 Quiz

Wk	Date	Topic
10	Oct 26	Unit 4 begins
11	Nov 2	Unit 4 (cont.)
12	Nov 9	Unit 4 (cont.)
13	Nov 16	Unit 4 (cont.) — Unit 4 Quiz
14	Nov 23	Unit 5 begins
15	Nov 30	Unit 5 (cont.)
16	Dec 7	Unit 5 (cont.) — Unit 5 Quiz
17	Dec 16	FINAL EXAM

Tentative schedule — subject to change. Updates will be announced by the instructor.

SLIDE 6 *Course Timeline*

NOTES

Generative AI Policy

Academic Integrity

Participation Verification

Title IX

Leeward CC prohibits sex- and gender-based discrimination. Contact the Title IX Coordinator at leetix@hawaii.edu for resources and support.

NOTES

Specimen Schedule — What You'll Examine Each Week

Wk	Topic	Alloys Examined	Wk	Topic	Alloys Examined
1	Atomic Structure	1018 steel (new)	10	Wrought Aluminum	6061-T6, 2024-T3 Al (new: 2024-T3)
2	Mech. Properties	1018, 1045 steel (new: 1045)	11	Aluminum Forming	6061-T6, 5086 Al (new: 5086)
3	NO CLASS	—	12	Machining Processes	1018, 1045 steel, 6061-T6 Al (revisit)
4	Carbon & Alloy Steels	1018, 1045, 1075 steel (new: 1075)	13	Copper Alloys	110, 182 Cu (new)
5	Stainless Steel	304, 316, 17-4PH SS (new)	14	Brass Alloys	260, 360, 464 brass (new: 464)
6	Nonferrous Metals	6061 Al, 260 brass (new)	15	Precip. Hardening	17-4PH SS, 2024-T3 Al, 182 Cu (revisit)
7	Mech. Testing	1018, 1045 steel (revisit)	16	Capstone & Synthesis	— (synthesis)
8	NDT Methods	1018 steel, 360 brass, Ti-6Al-4V (new: 360, Ti)			
9	Quality & Defects	1018 steel (revisit)			

Each week's metallography module draws from the shared 15-specimen kit. "Revisit" means an alloy you've already examined, now viewed through a new lens.

SLIDE 10 Specimen Schedule — What You'll Examine Each Week

NOTES

